MLSC AIML Domain Lead Recruitment Challenge

Project: MLSC Knowledge Assistant

Problem Statement

Build an AI-powered knowledge assistant for MLSC that can answer user questions using the provided MLSC knowledge base. The system should retrieve relevant information from the database and generate accurate, grounded answers.

The assistant should also be able to recognize when the required information is not available in the provided knowledge base and avoid making up an answer.

Provided Knowledge Base

You will receive a folder containing sample MLSC documents. The information may be distributed across multiple files, and some questions may require information from more than one document.

`mlsc_knowledge_base/`

File	Contains
about_mlsc.txt	General information about MLSC
domains.txt	MLSC technical domains
leadership.txt	Leadership structure and responsibilities
membership.txt	Membership and coordinator information
hackathons.txt	Hackathon-related information
code_of_conduct.txt	Community guidelines

Core Requirements

Your solution must:

- Allow a user to ask questions in natural language.
- Use the provided knowledge base to answer questions.
- Provide the relevant source document(s) for each answer.
- Handle questions whose answers require information from multiple documents.
- Handle questions for which the answer is not present in the knowledge base.
- Include an evaluation component for measuring the quality of the system.

Evaluation

You will be provided with a separate evaluation set containing questions and reference answers. Use it to evaluate your system and report meaningful results.

Your evaluation should consider, at minimum:

- Context Precision / Retrieval Precision
- Context Recall / Retrieval Recall
- Answer Relevancy

- Faithfulness / Groundedness

You may use an evaluation framework such as RAGAS, DeepEval, or another suitable approach. You are expected to understand and justify the metrics you report.

Sample Question Types

The evaluation set may include questions such as:

- Direct questions — e.g. What technical domains exist in MLSC?
- Multi-document questions — requiring information from multiple files.
- Reasoning questions — requiring synthesis of information.
- Unanswerable questions — where the correct behavior is to say that the information is unavailable.
- Ambiguous or differently worded questions.

Deliverables

1. Working application / demo.
2. Source code in a GitHub repository.
3. README containing setup and run instructions.
4. Brief explanation of your approach and major technical decisions.
5. Evaluation results with the metrics you selected.

Technology

You may choose your own programming language, LLM/API, embedding model, vector database, framework and UI technology. You are not required to use any particular stack.

Important Constraints

- Do not hard-code answers for the supplied questions.
- The provided knowledge base must be the source of truth for MLSC-specific information.
- AI coding assistants may be used, but you must be able to explain your implementation during the interview.
- Focus on correctness and system quality rather than an elaborate UI.

Time Limit

Recommended working time: 6 hours.

The interview will include a short demonstration and technical discussion of your implementation, design decisions and evaluation results.

What We Will Look For

The challenge is intended to assess your ability to independently design and reason about an AI system. We will consider retrieval quality, answer quality, evaluation methodology, handling of unsupported questions, technical decisions, code quality and your ability to explain the system.